

IDENTIFICATION OF ICE FLOE IMPACTS IN AN INERTIAL TIME SERIES AND IMPLICATIONS FOR WAVE MEASUREMENT IN PANCAKE ICE

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ABSTRACT

Six ice-tethered instruments were deployed in the Antarctic Marginal Ice Zone during July 2022, coinciding with the movement of a polar cyclone through this region. High frequency inertial and GPS location data were retrieved from the instruments. Inertial data were analysed from two of the instruments to investigate the waves-in-ice dynamics of the ice floes on which the instruments were deployed. In particular, large frequent impacts between floes were detected in the inertial time-series. This in turn has implications for wave-ice interaction and attenuation, wave measurement and calibration of remote sensing data.

Index Terms— Sea ice, Antarctic, Marginal Ice Zone, Inertial Measurement Unit, waves-in-ice

1. INTRODUCTION

The Antarctic Marginal Ice Zone (MIZ) is a transitional region between open ocean and consolidated pack ice [1]. The region is of key scientific interest as the climatic processes in the region, in particular the sea ice extent, have implications for the global climate [2, 3]. The high variability and heterogeneous nature of sea ice makes its difficult to model [4]. In-situ measurement of sea ice dynamics, particularly the interaction of waves and sea ice, known as waves-in-ice is necessary to further develop models of this region [5, 6] and calibrate remote sensing devices [7, 8].

In recent years, several in-situ ice-tethered devices have been developed [9, 8, 5] to measure waves-in-ice interactions in the Antarctic MIZ. However, due to the difficulty in retrieving the devices as well as the cost of sending raw inertial data via satellite link, these transmit summary wave statistics, which calculate the spectral moments of a power spectrum of a 15 to 30 minute measurement period.

As the acceleration measurements of the ice-floe, on which these devices are deployed, are taken as a proxy measurement of the underlying wave activity, the dynamics of the

ice floe could potentially bias the wave measurement in cases where the floe diameter is large relative to the incoming wave length. The long wave period is assumed make the dynamics of the floe negligible for wave measurement as the it provides sufficient time for the response amplitude operator (a measure of the response amplitude of a floating body relative to the amplitude of an exciting wave) of the floe relative to the wave to tend to 1 as long as the individual floe diameter is short relative to the length of the wave [10].

Ice floe collisions are a known Antarctic MIZ phenomenon, which in the simplest case is driven by wave orbitals. Floe collisions have been investigated extensively in wave tank settings [11] and have been studied in the Antarctic MIZ [12]. They are of interest in and of themselves due to their potential as a dissipative mechanism for wave energy contributing to wave attenuation [4, 12]. It was noted that they have the potential to bias wave measurement during Arctic in-situ experimental deployments where the raw data were remotely transmitted [13].

2. ICE TETHERED DEVICES

These devices were constructed to collect both in-situ data and survive harsh Antarctic conditions. To achieve this a housing consisting of an microcontroller unit, IMU, GNSS and Iridium modem were enclosed in a plastic housing and placed in a raised stand to prevent them from freezing into the ice. The stand was fixed to the ice by the use of metal spikes. The implication of a raised stand for IMU measurement were accounted for by determining the angular acceleration relative to the base of the stand. Each ice-tethered buoy collected and stored high frequency inertial time-series data of sea ice motion (3-axis linear acceleration, 3-axis rotational velocity at ≥ 100 Hz for 15-minute periods) as well as positional data (GNSS) to measure ice drift.

The onboard IMU was an InvenSense ICM20649 sampled at 104 Hz. The sampling frequency was chosen such that it would exceed the Nyquist criterion for wave measurement in the MIZ which experiences a wave period of 5 to 20 seconds

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[9]. Additionally, the high sampling frequency allows for filtering and identification of high frequency phenomena such as floe-floe collisions. Due to the magnetic distortion close to the polar regions, magnetometer data were not collected and directional wave measurements not determined.

3. DEPLOYMENT

Six ice-tethered "SHARC" buoys, developed by the University of Cape Town (UCT), were deployed in the Antarctic Marginal Ice Zone over a spatially distributed region (each buoy separated by 25 km) during the Southern Ocean Seasonal Experiment (SCALE-WIN22) Winter 2022 campaign (19 July to 24 July 2022). Buoy deployment position and retrieval time were optimised to measure waves-in-ice and ice drift effects of a polar cyclone moving through the MIZ region during the experimental period. Buoys were placed in open drift to consolidated ice conditions to measure changes in significant wave height, wave period and power spectrum. Each device was deployed close to the centre of a pancake ice floe, which were fused together into a consolidated sheet. Data were stored locally, processed onboard and summary wave statistics (significant wave height, wave period and power spectrum) were transmitted every 40 minutes via the Iridium satellite network. Four UCT buoys were retrieved at the end of the experimental period (up to 5 days from deployment) and the onboard data were recovered.

During the course of the deployment the large consolidated ice floes on which the buoys were deployed fractured into the composite pancake ice floes due to mechanical breakup and change in atmospheric temperature.

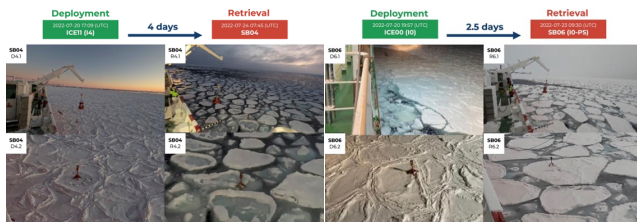


Fig. 1. Deployment and retrieval of SHARC Buoy 04 (SB04) (left) and SB06 (right). Photo credit: A. Spirakis

The devices were collected off the ice and the onboard high frequency datasets retrieved.

4. WAVES-IN-ICE MEASUREMENTS

The wave statistics were extracted using a standard approach. Firstly, segmenting the data into 30 minute samples. Secondly, determining the vertical acceleration and integrating from acceleration to displacement in the frequency domain [9, 8]. Thirdly, performing Welch's method to calculate the power spectrum. Finally, the spectral moments were then

calculated, allowing the calculation of the significant wave height and the mean wave period.

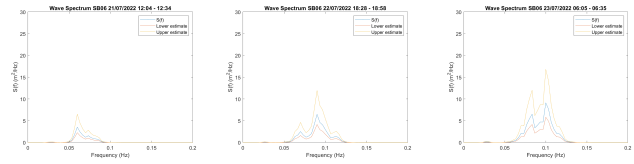


Fig. 2. Development of the wave spectrum measured by SB06. Plots are selected to show before, during and immediately after the polar cyclone passed into the region.

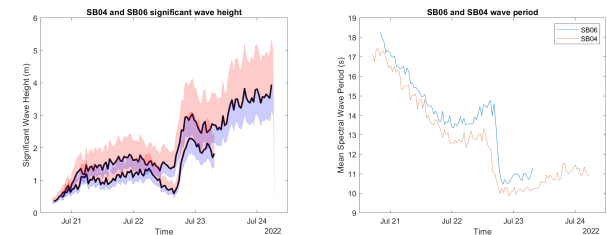


Fig. 3. Significant wave height (left) and mean wave period (right) during the course of the deployment. A 90% certainty interval is shown for the significant wave height. On the right figure SB06 is the lower plot.

The polar cyclone that coincided with the deployment on the evening can be clearly seen in fig. 3, resulting in a sharp increase in wave height and a corresponding decrease in mean wave period as the higher frequency waves generated from the cyclone entered into the region which was previously dominated by lower frequency swell as shown in figure 2.

5. FLOE-FLOE COLLISIONS

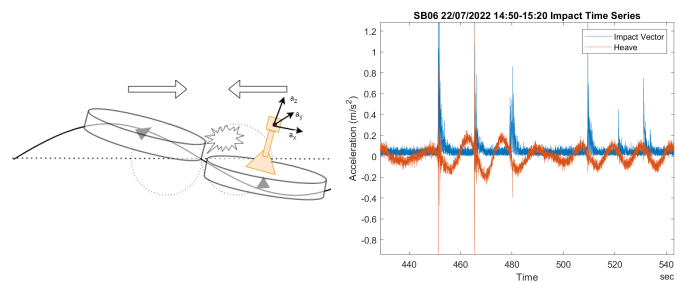


Fig. 4. The collisions experienced by the buoys were identified by the comparison of acceleration in the horizontal plane relative to the vertical acceleration.

The collisions between floes were identified by comparing the vertical acceleration, to the acceleration in the plane parallel to the ocean surface (see 4). The acceleration spikes are hypothesized to be collisions based on wave tank testing

studies [14, 11]. A high pass filter with a cutoff frequency of 10 Hz was implemented to remove any underlying wave acceleration artifacts. If acceleration in the parallel plane exceeded the acceleration 4 standard deviations from the mean in the vertical direction it was considered to be a wave impact [11]. Additionally, a peak-finding algorithm with a minimum distance between peaks of 0.1 seconds was used to count the collisions, allowing a differentiation between impacts.

These impacts were then quantified as impacts per wave period to determine the relationship, if any, between the two. Before the arrival of the wave field from the polar cyclone, these collisions correlated to the wave period with several impacts per wave period corresponding to the wave tank experiments investigating collisions driven by orbital wave motion [11, 14]. It can be seen in figure 5 that there is no clear relationship after the arrival of the wave field from the polar cyclone. However, it provides a normalized comparison of collision frequency to compare the two devices over time.

It is clear that there is a substantial increase in collision events as soon as the polar cyclone moved through the region on the 22 July 2022. The extremely high number of impacts for SB06 (~70 per wave period) during the arrival of the storm is hypothesised to be as a result of the breakup of the ice field.

This then leads to a concern that the sheer number of collisions during this period of interest could lead to biasing of the statistical wave measurement during a sampling period of particular interest.

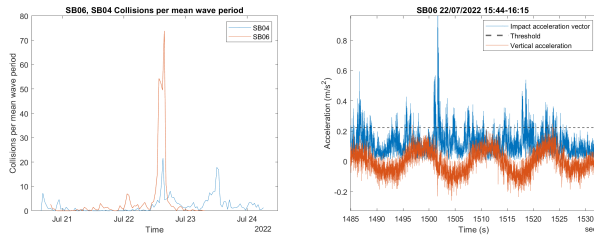


Fig. 5. Measured collisions normalized per wave period for SB04 and SB06 (left). The high incidence of acceleration spikes (collisions) shown relative to the vertical acceleration with the acceleration threshold used to count the collisions shown (right).

6. BIASING OF WAVE MEASUREMENTS

There are generally two approaches to determining the vertical acceleration of a floe used by ice-tethered instruments. The first is to use a small angle assumption that due to the long wavelength of incoming waves, the tilt of the floe is significantly less than 10° and so the heave is equivalent to the vertical acceleration [9], $\hat{\mathbf{A}}_z^i \approx \hat{\mathbf{A}}_z^b$. The second is to fuse the measurements from all axis of the IMU: roll (ϕ , pitch (θ)),

heave ($\hat{\mathbf{A}}_z^b$), surge ($\hat{\mathbf{A}}_x^b$) and sway ($\hat{\mathbf{A}}_y^b$) using the following equation [9, 8]:

$$\hat{\mathbf{A}}_z^i = \mathbf{g} - (\sin(\theta)\hat{\mathbf{A}}_x^b + \sin(\phi)\cos(\theta)\hat{\mathbf{A}}_y^b + \cos(\phi)\cos(\theta)\hat{\mathbf{A}}_z^b) \quad (1)$$

What is important to note is that the surge and sway components of acceleration are included in the calculation. Significant changes in these measurements, particularly if the platform is tilted greater than 10° will be included in the wave measurements.

The IMU measurements are passed through a Kalman filter to estimate the orientation of the platform and then equation 1 is used.

To determine whether the impacts affect the signal, the input signal was low pass filtered with a 2 Hz cutoff frequency. Then the significant wave height estimates are then compared to wave measurements without pre-filtered data.

Fortunately, no biasing results from the impacts. This is because the high number of impacts coincided with relatively low wave height as can be seen by comparing the time at which the collisions increased in number to the time at which the wave height increased. (fig. 3 vs fig. 5). The result is a low platform tilt and so the horizontal acceleration components are negligible.

However, When the the higher frequency waves moved into the region and the significant wave height increased, the waves became steeper resulting in a higher platform tilt. Instruments that are equipped to use a single accelerometer [9], generally with higher accuracy than a low cost IMU, using the small angle assumption are then in this case susceptible to some measurement bias as can be seen by comparing the wave measurement derived by heave to the wave measurement derived by fusing information from all axes (fig. 6) This may be of concern due to the increasing penetration of waves-in-ice [4].

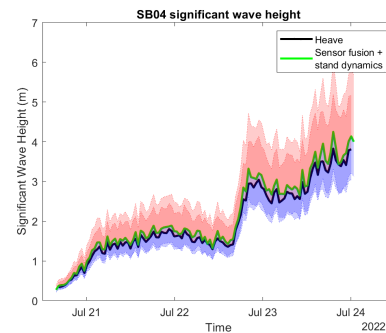


Fig. 6. Wave measurement using the small angle assumption (heave only) measurement compared to a sensor fusion approach for instrument SB04

7. CONCLUSIONS

A novel dataset showing the collisions between ice floes measured by an ice-tethered instrument in the Antarctic was presented.

The incidence of collisions, identified by using an acceleration threshold after pre-processing to distinguish wave activity from floe-floe collisions was shown. Additionally, these collisions coincided with the breakup of a consolidated ice field.

We found that the collisions did not contribute to biasing of the measurement of uni-dimensional significant wave height. However, instruments that use heave measurement only will suffer from some bias in steeper waves (fig. 6). This is increasingly likely as these sensors were deployed on consolidated ice in the winter. Thus it is suggested that wave measurements should use sensor fusion (generally Kalman filtering) and can as a precaution lowpass and decimate the raw data (such as in [5]).

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